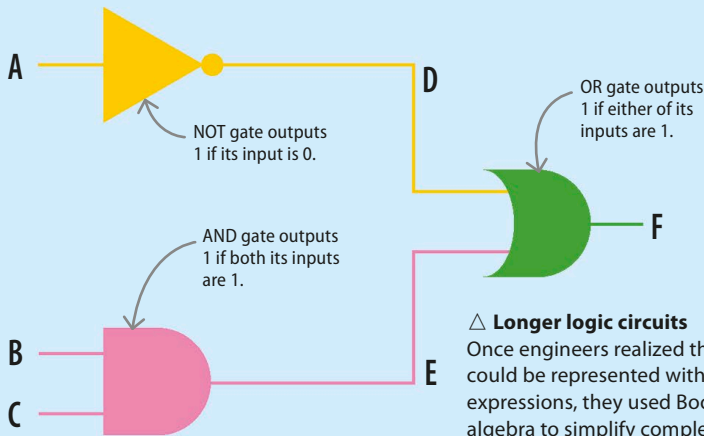


Logic expressions

The AND, OR, and NOT operators used in Boolean logic correspond to the AND, OR, and NOT logic gates used in circuit design. Several logic gates can be connected together to make a logic circuit. Logic circuits can be designed to calculate anything from simple arithmetic to the physics of spaceship orbits. Understanding Boolean algebra helps programmers build and test these abstract circuits.

A	B	C	D	E	F
			NOT A	B AND C	D OR E
0	0	0	1	0	1
0	0	1	1	0	1
0	1	0	1	0	1
0	1	1	1	1	1
1	0	0	0	0	0
1	0	1	0	0	0
1	1	0	0	0	0



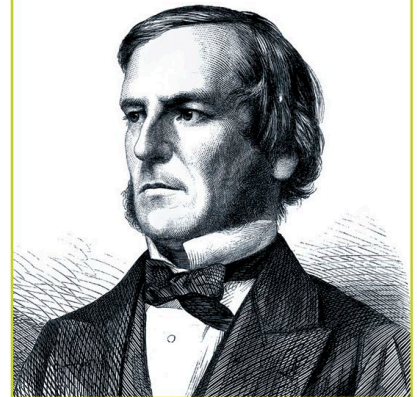
△ Longer logic circuits

Once engineers realized that circuits could be represented with Boolean expressions, they used Boolean algebra to simplify complex circuits.

BIOGRAPHY

George Boole

An influential mathematician, George Boole (1815–1864) didn't have much access to formal education and was mostly self-taught. He invented Boolean logic as a systematic, mathematical approach to ancient Greek philosopher Aristotle's theory of categorical logic – rules used to determine if a statement is true or false. Boole explained his concept of Boolean logic in his work titled *The Laws of Thought* (1854).



Logical search

Boolean logic can also be used to search databases quickly. It eliminates the need for going through an entire database by making each search more precise. By combining keywords – such as dessert – with Boolean operators, a database search can take seconds.

▷ Boolean query

A Boolean query uses Boolean operators to create a more effective search by excluding irrelevant possibilities.

